CUSTOMER SEGMENTATION USING CLUSTERING

PROJECT

Presented by:AnuskaVerma,Rajeev,
Devraj gadhavi,Diya kumari.

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01 ABSTRACT

This project focuses on customer segmentation using multiple clustering techniques to uncover distinct customer groups and support targeted marketing strategies. The dataset was cleaned, preprocessed, and scaled to ensure quality and uniformity. Extensive Exploratory Data Analysis (EDA) was conducted using boxplots, histograms, countplots, correlation heatmaps, and monthly sales trends to identify outliers, patterns, and relationships within the data. Several clustering methods, including K-Means and alternative algorithms, were applied to segment customers effectively. The optimal cluster numbers were determined using metrics such as the Elbow Method and Silhouette Score, with Principal Component Analysis (PCA) used for visualization. The resulting segments were analyzed and profiled to highlight purchasing behaviors, spending levels, and common traits. Finally, tailored marketing strategies were recommended for each cluster, offering actionable insights to enhance customer engagement and business decision-making.

02 INTRODUCTION

In today's highly competitive business environment, understanding customer behavior is essential for designing effective marketing strategies and improving customer retention. Businesses often deal with large volumes of customer transaction data, which can provide valuable insights if analyzed properly. However, this data is often unstructured, noisy, and complex, making it difficult to extract actionable knowledge using traditional approaches.

Customer segmentation is a widely used data-driven technique that allows businesses to group customers with similar characteristics, behaviors, or preferences. By analyzing these segments, organizations can tailor marketing campaigns, optimize product offerings, and enhance overall customer satisfaction. Among the many techniques available, clustering algorithms such as K-Means, Hierarchical Clustering, and DBSCAN are particularly useful because they group customers without requiring predefined labels.

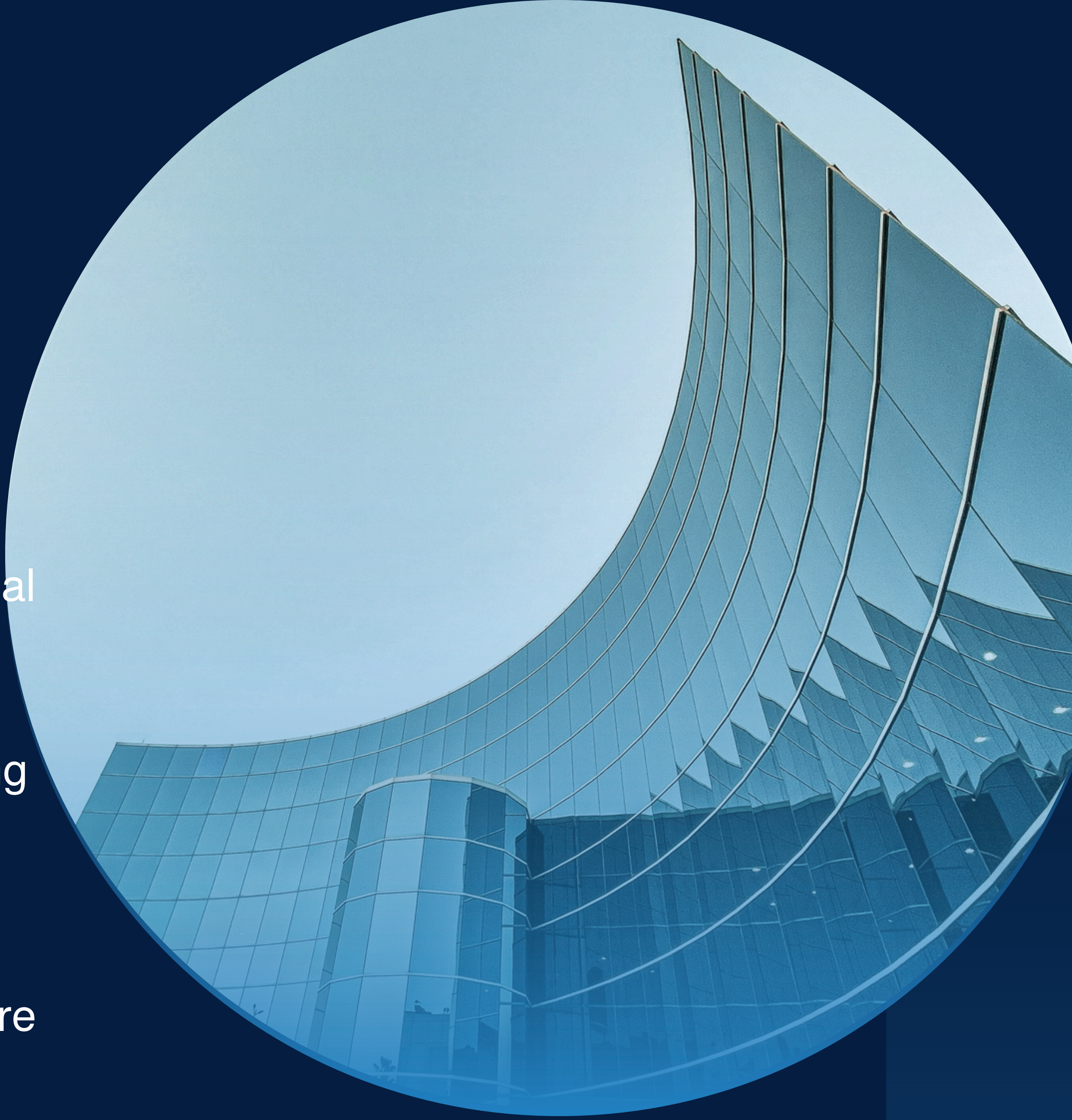
This project aims to perform customer segmentation using a real-world transactional dataset. The process began with data cleaning, preprocessing, and scaling, ensuring that the dataset was free of inconsistencies and ready for analysis. Exploratory Data Analysis (EDA) was conducted to better understand customer demographics and purchasing behavior, using a variety of visualizations such as boxplots, histograms, countplots, and correlation heatmaps. After preparing the data, multiple clustering methods were applied, with the Elbow Method, Silhouette Score, and other evaluation metrics used to determine the optimal number of clusters. Dimensionality reduction techniques like Principal Component Analysis (PCA) were further employed to visualize and interpret the clusters effectively.

The insights derived from clustering were then used to profile each customer segment in terms of spending behavior, frequency of purchase, and product preferences. Based on these profiles, targeted marketing strategies were recommended for each cluster, enabling businesses to make data-driven decisions that enhance customer engagement and revenue growth.

03 PRE-PROCESSING

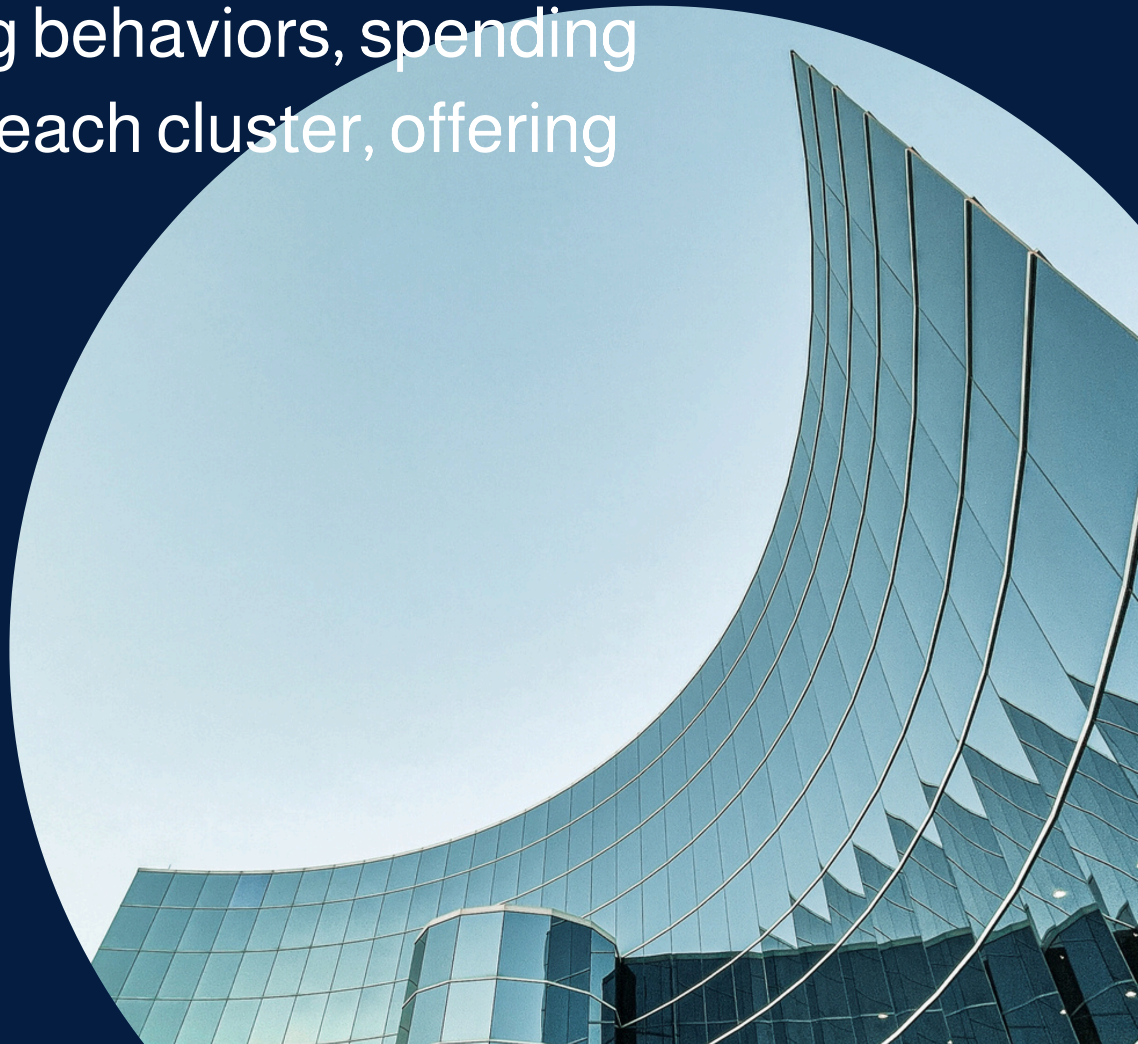
Preprocessing was a crucial step in preparing the dataset for clustering. The original dataset contained transactional details such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. To ensure the quality and reliability of the analysis, the following preprocessing steps were performed:

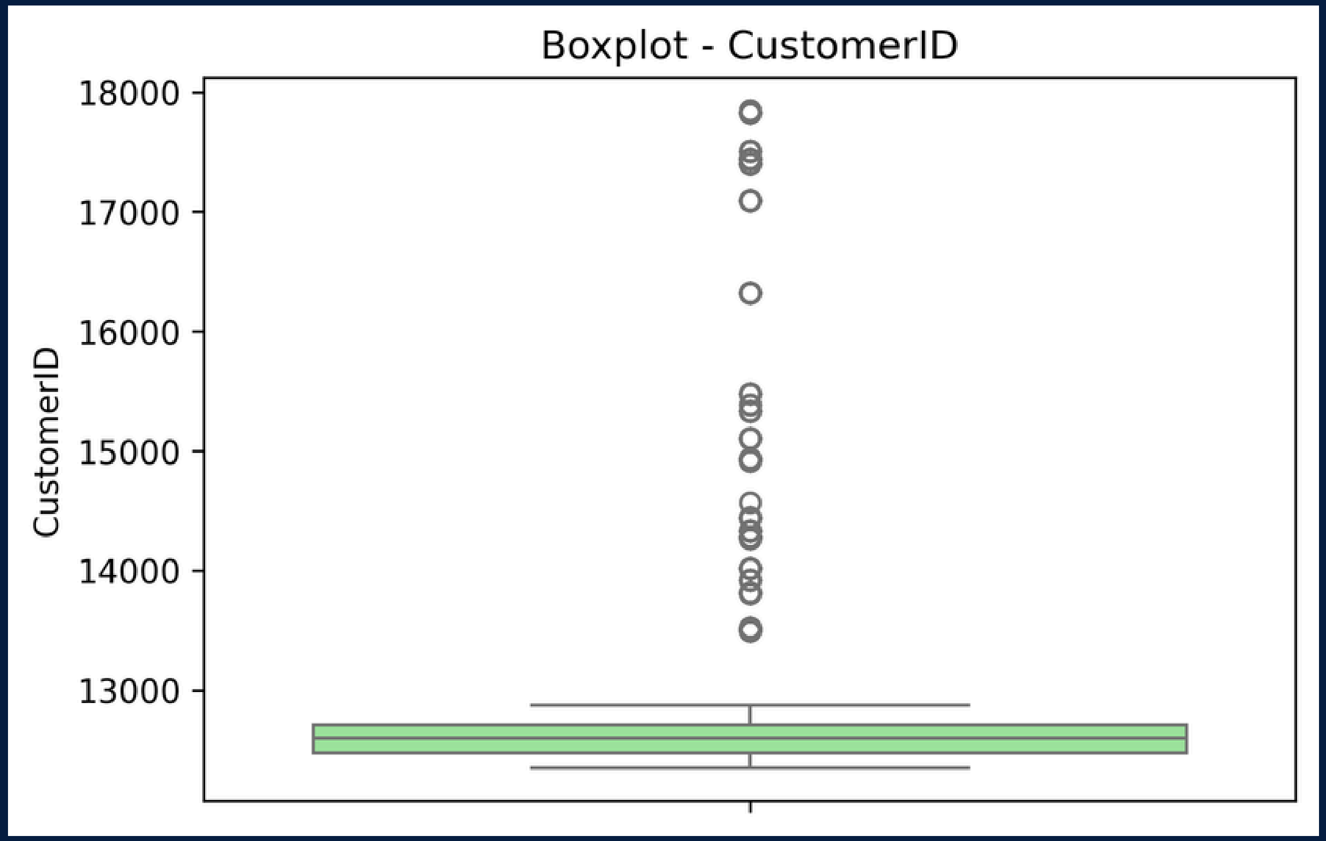
1. Handling Missing Values – Missing entries in critical fields like CustomerID and Description were identified using heatmaps and count checks, and rows with incomplete or invalid data were removed.
2. Removing Duplicates and Outliers – Duplicate records were dropped, while extreme values in Quantity and UnitPrice were detected using boxplots and handled to avoid skewing the results.
3. Data Cleaning – Irrelevant or incorrect records, such as negative quantities (returns) and zero-priced items, were filtered out.
4. Feature Engineering – New features such as monthly quantity and monthly revenue were derived to better capture customer purchasing behavior.
5. Data Transformation – Categorical attributes like Country were encoded where required, and date fields such as InvoiceDate were transformed to extract time-based insights.
6. Scaling – Since clustering algorithms like K-Means are sensitive to data magnitude, all numerical features were standardized using MinMaxScaler/StandardScaler, ensuring equal contribution of features during clustering.

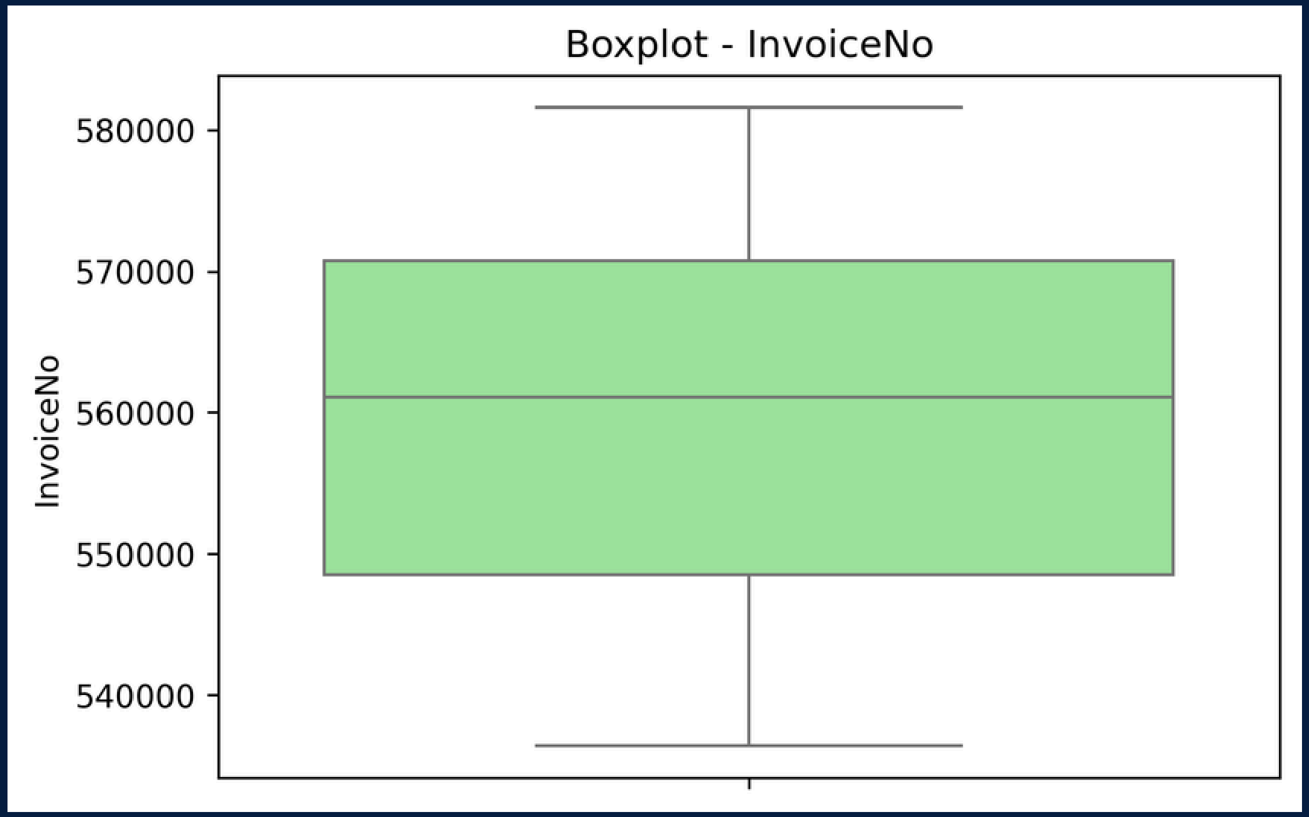


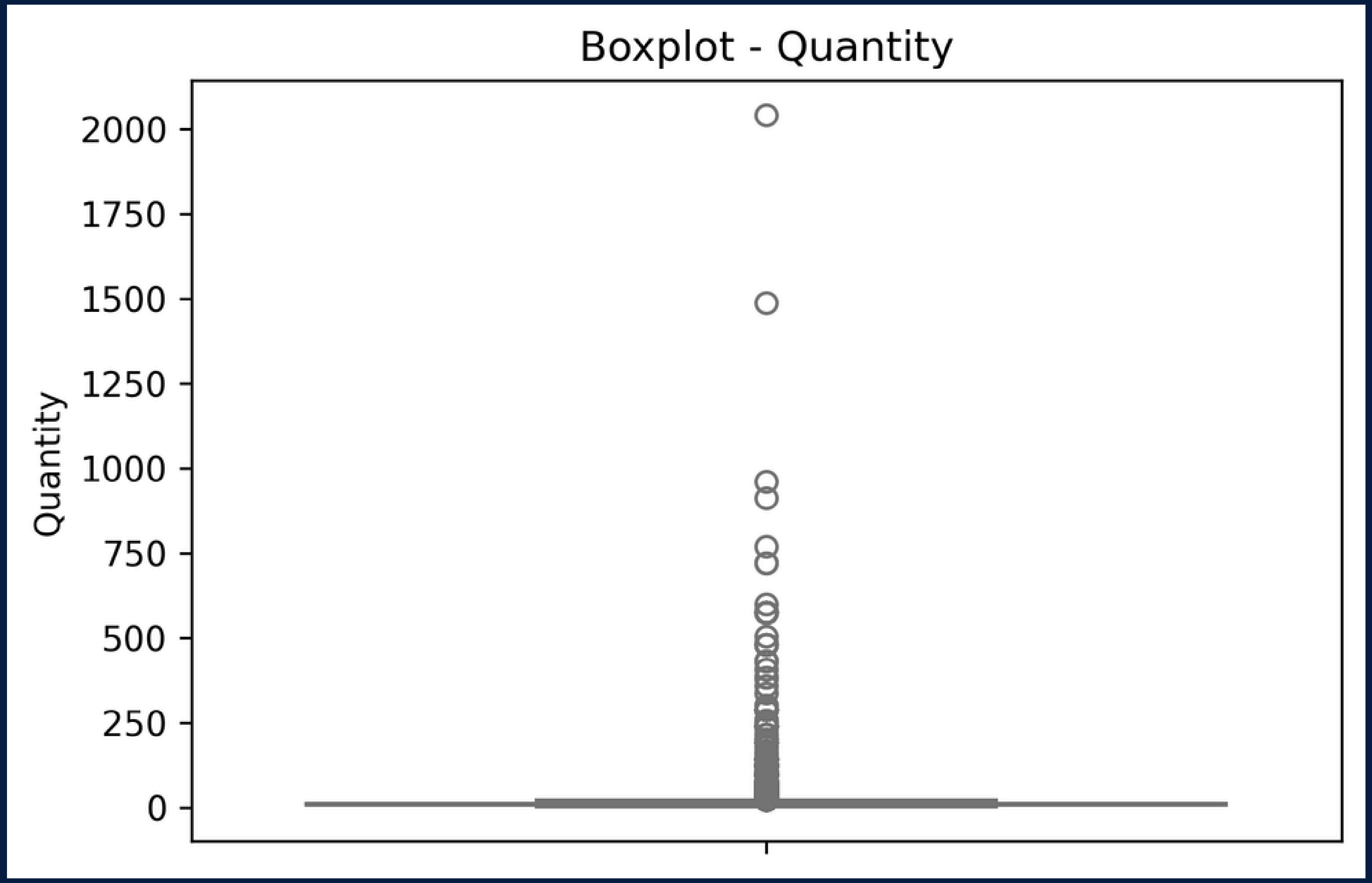
04 EXPLORATORY DATA-ANALYSIS

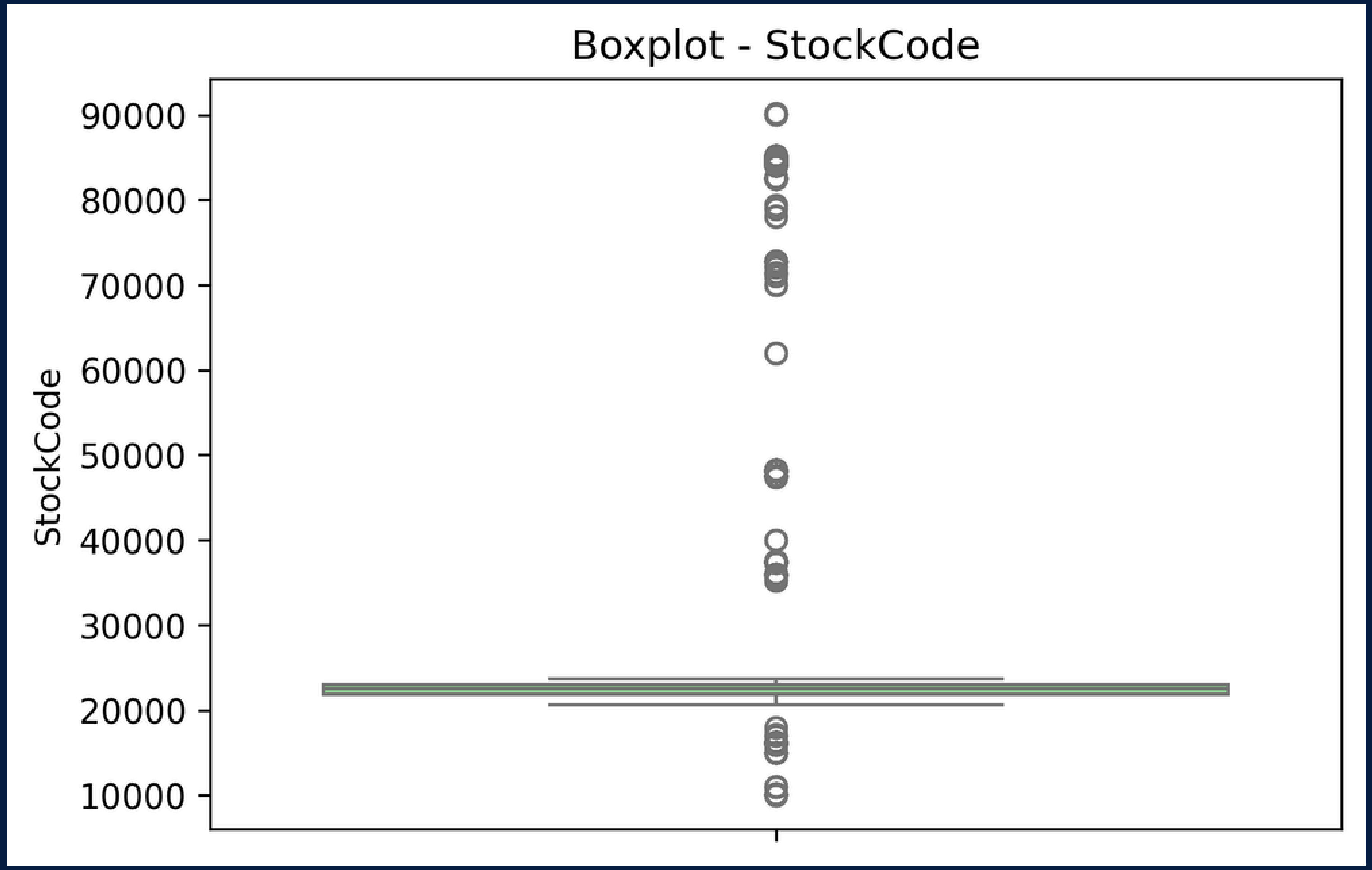
This project focuses on customer segmentation using multiple clustering techniques to uncover distinct customer groups and support targeted marketing strategies. The dataset was cleaned, preprocessed, and scaled to ensure quality and uniformity. Extensive Exploratory Data Analysis (EDA) was conducted using boxplots, histograms, countplots, correlation heatmaps, and monthly sales trends to identify outliers, patterns, and relationships within the data. Several clustering methods, including K-Means and alternative algorithms, were applied to segment customers effectively. The optimal cluster numbers were determined using metrics such as the Elbow Method and Silhouette Score, with Principal Component Analysis (PCA) used for visualization. The resulting segments were analyzed and profiled to highlight purchasing behaviors, spending levels, and common traits. Finally, tailored marketing strategies were recommended for each cluster, offering actionable insights to enhance customer engagement and business decision-making.

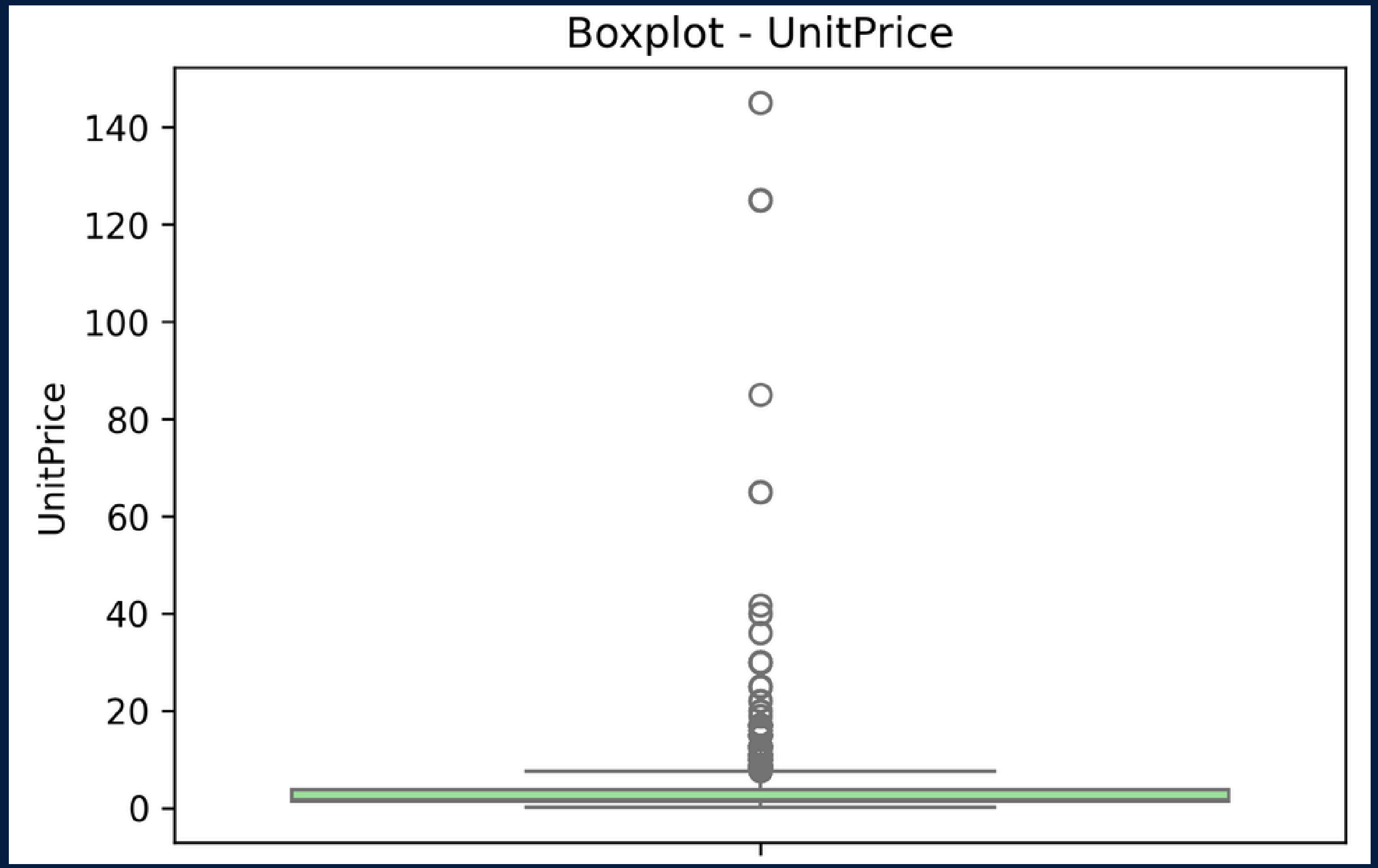


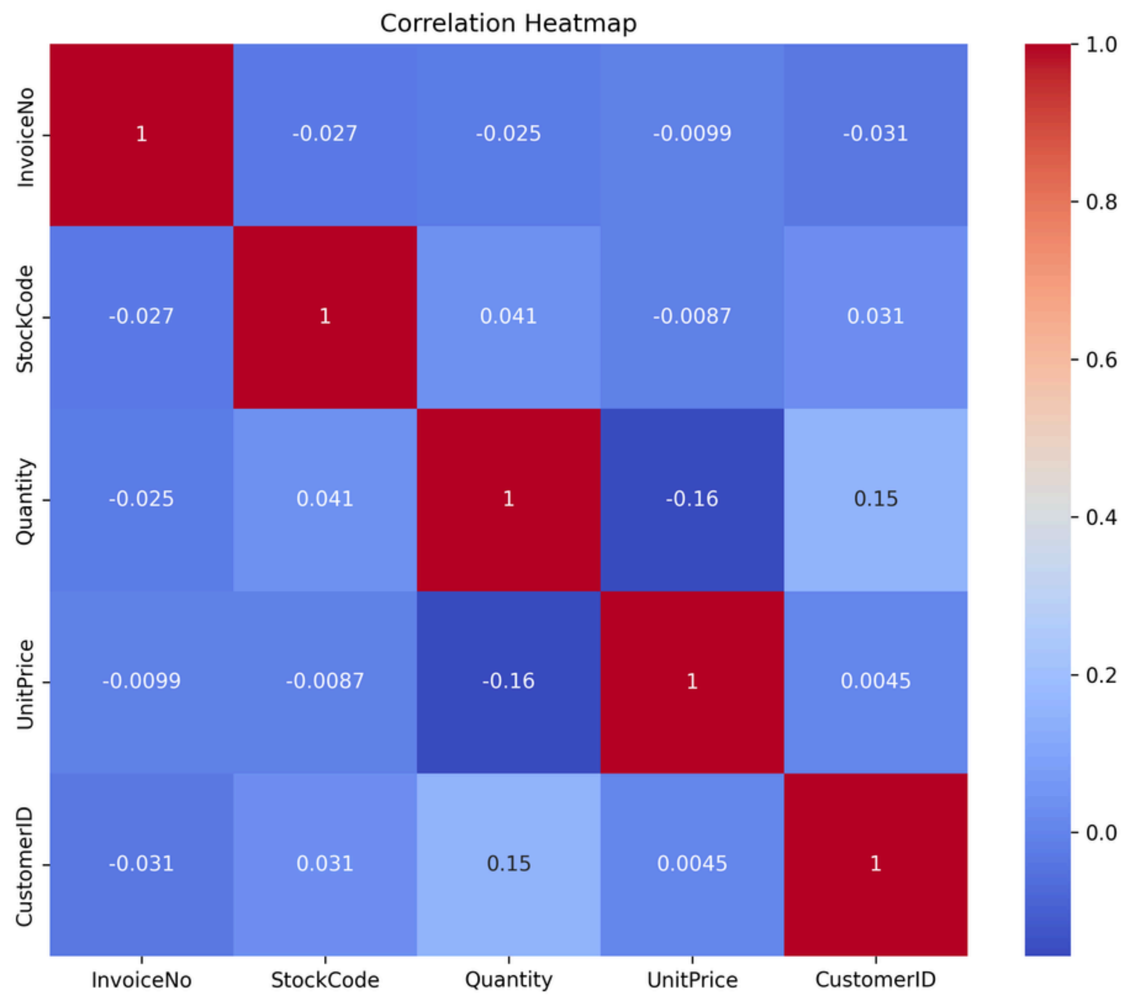

Boxplot - CustomerID
18000
17000
16000
15000
14000
13000
CustomerID

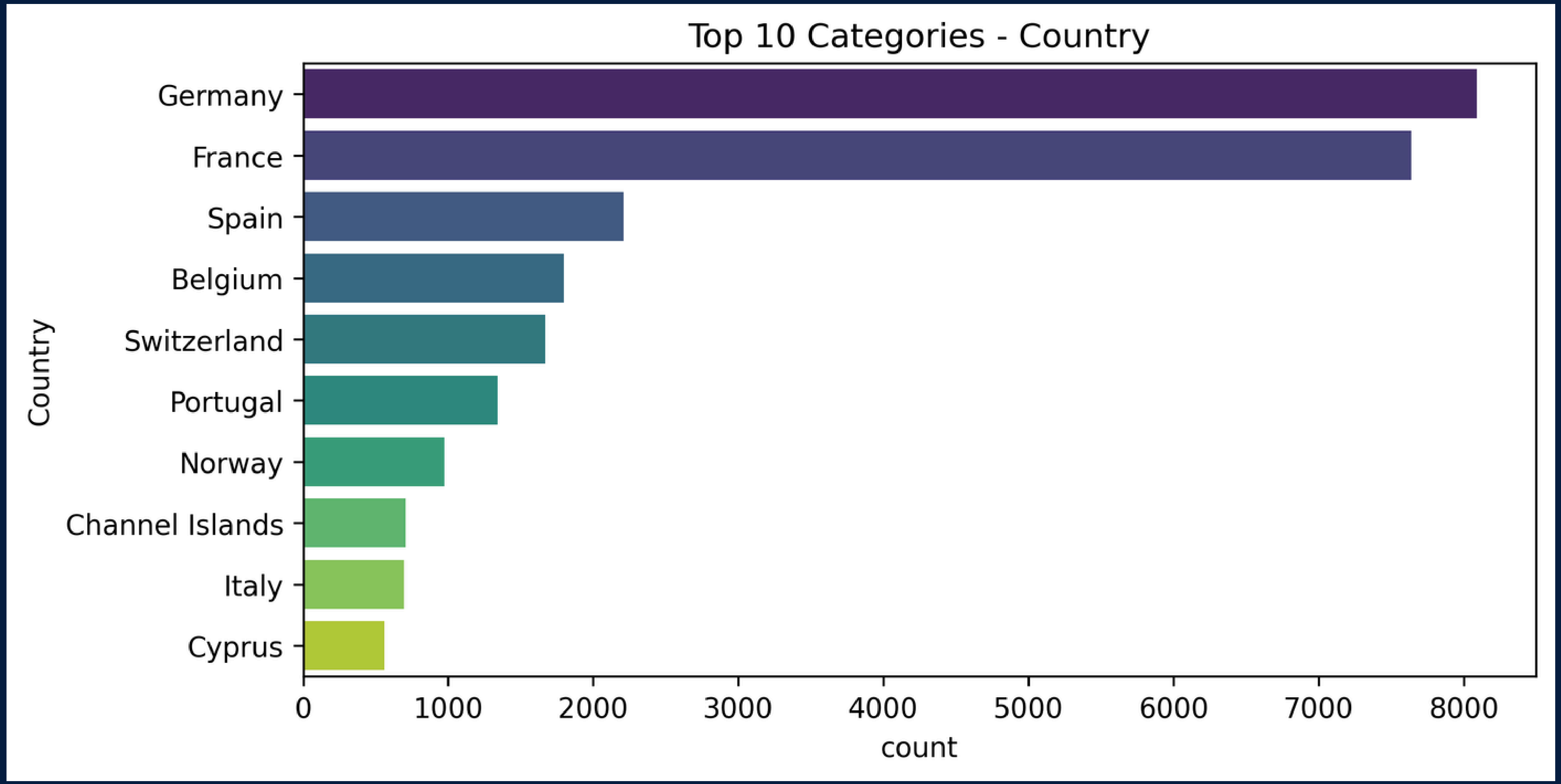

Boxplot - InvoiceNo
580000
570000
560000
550000
540000
InvoiceNo

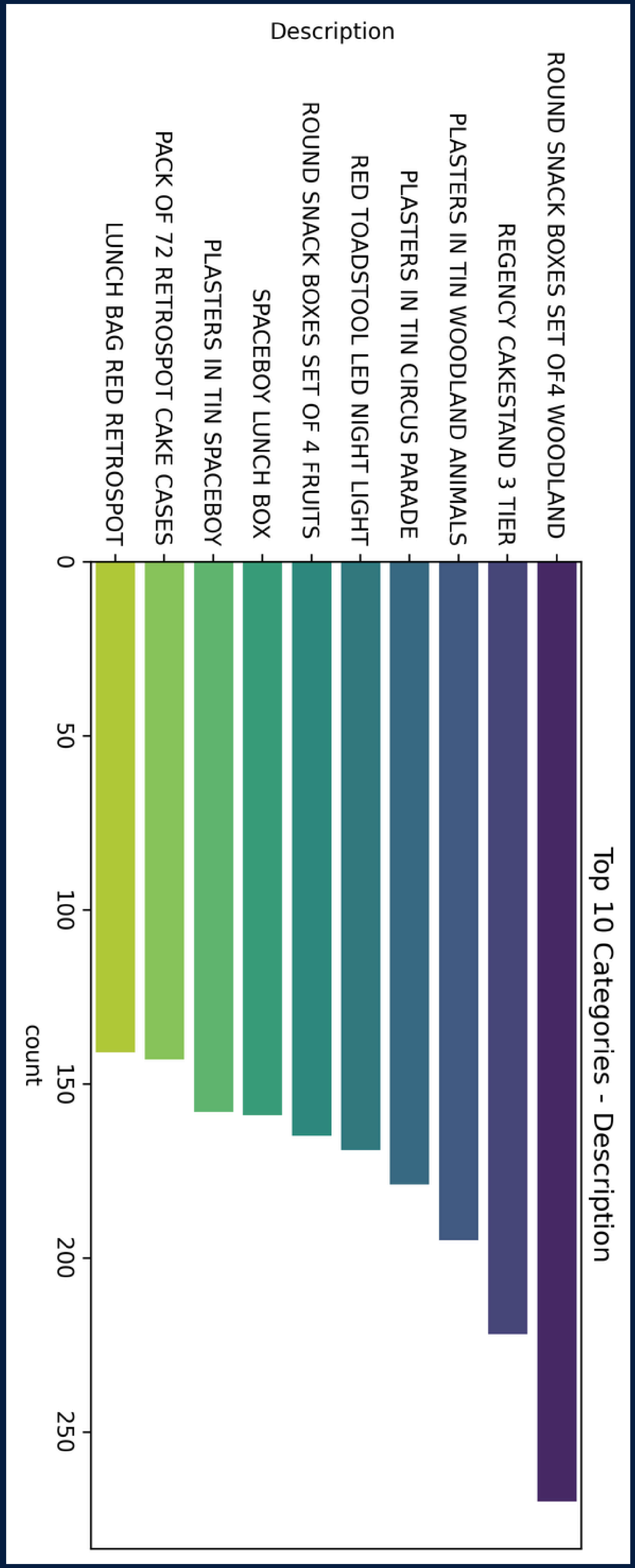

Boxplot - Quantity
2000
1750
1500
1250
1000
750
500
250
0
Quantity

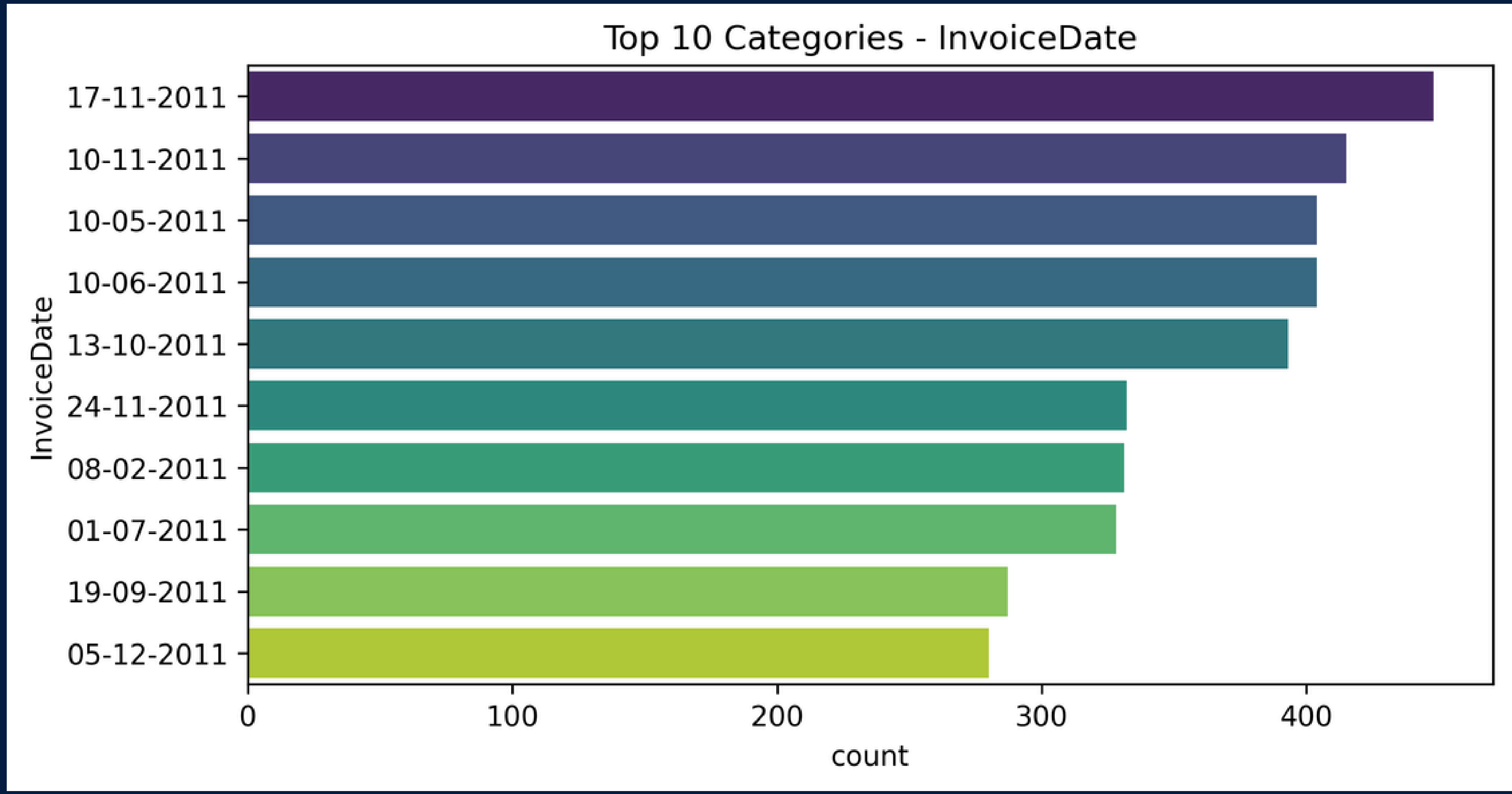

Boxplot - StockCode
90000
80000
70000
60000
50000
40000
30000
20000
10000
StockCode

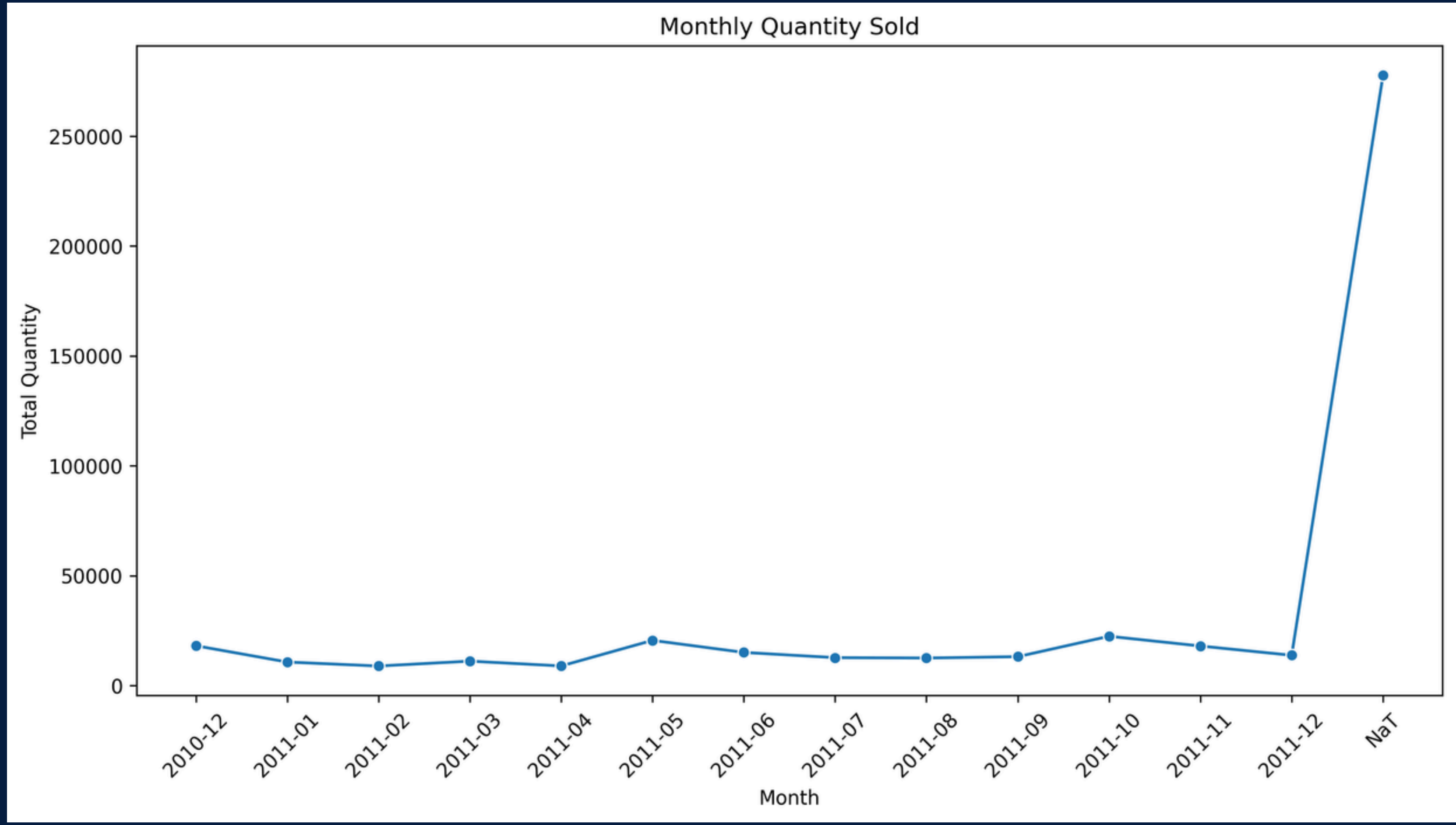

Boxplot - UnitPrice
UnitPrice
140
120
100
80
60
40
20
0

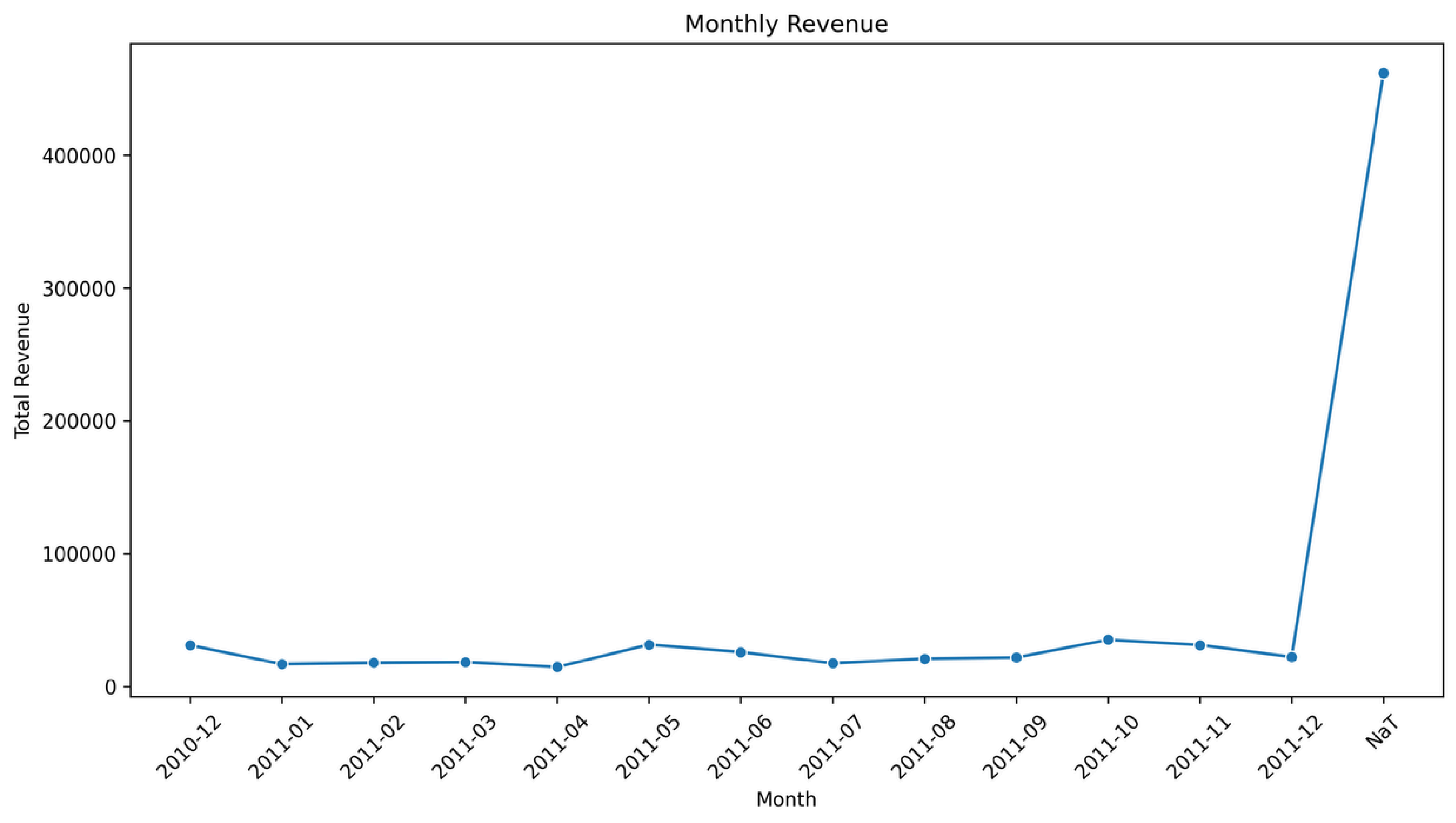

Correlation Heatmap
InvoiceNo
StockCode
Quantity
UnitPrice
CustomerID
1
-0.027
-0.025
-0.0099
-0.031
-0.027
1
0.041
-0.0087
0.031
-0.025
0.041
1
-0.16
0.15
-0.0099
-0.0087
-0.16
1
0.0045
-0.031
0.031
0.15
0.0045
1
1.0
0.8
0.6
0.4
0.2
0.0

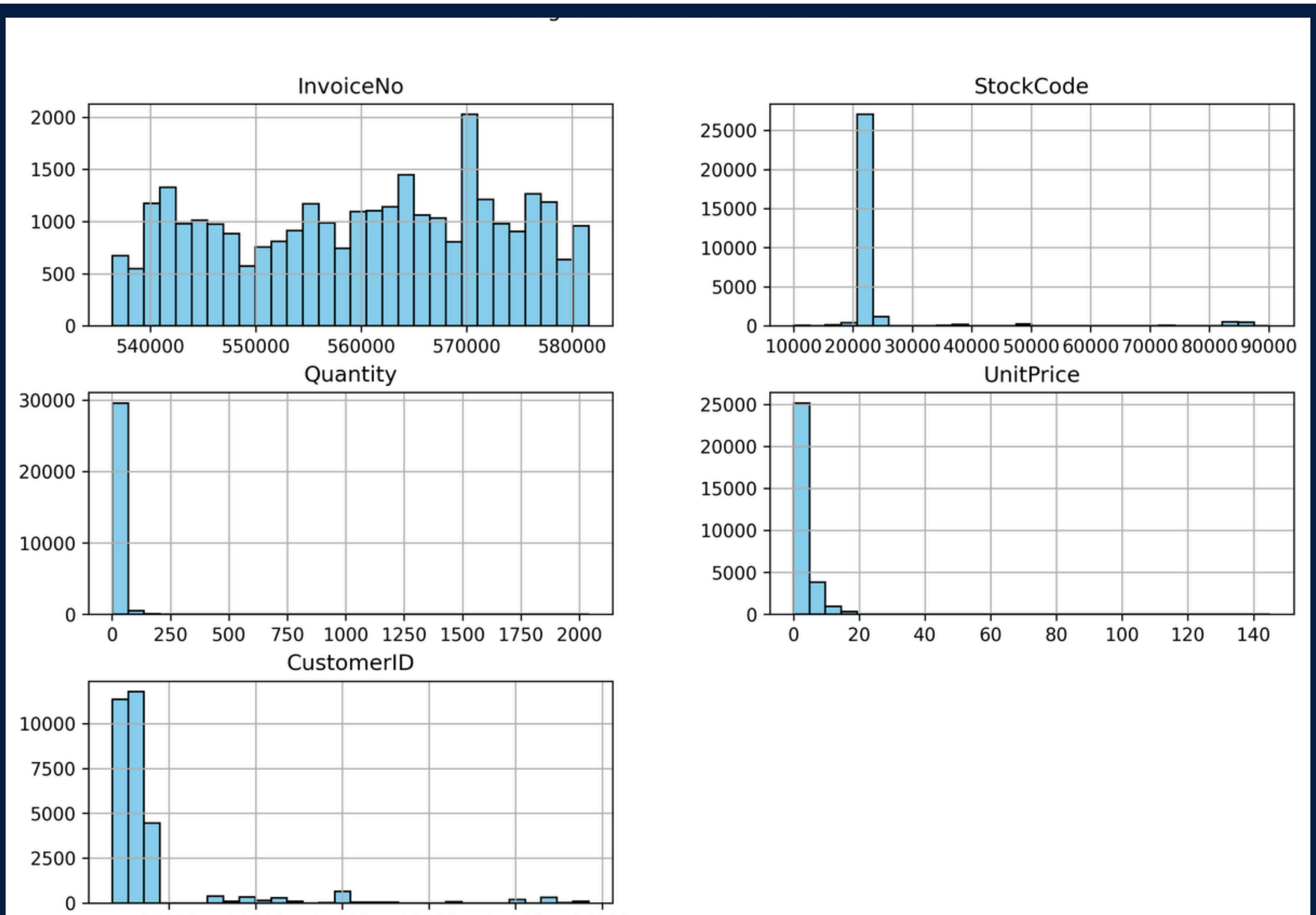

Top 10 Categories - Country
Germany
France
Spain
Belgium
Switzerland
Portugal
Norway
Channel Islands
Italy
Cyprus
Country
0
1000
2000
3000
4000
5000
6000
7000
8000
count


Description
LUNCH BAG RED RETROSPOT
PACK OF 72 RETROSPOT CAKE CASES
PLASTERS IN TIN SPACEBOY
SPACEBOY LUNCH BOX
ROUND SNACK BOXES SET OF 4 FRUITS
RED TOADSTOOL LED NIGHT LIGHT
PLASTERS IN TIN CIRCUS PARADE
PLASTERS IN TIN WOODLAND ANIMALS
REGENCY CAKESTAND 3 TIER
ROUND SNACK BOXES SET OF4 WOODLAND
0
50
100
150
200
250
count
Top 10 Categories - Description


Top 10 Categories - InvoiceDate
InvoiceDate
17-11-2011
10-11-2011
10-05-2011
10-06-2011
13-10-2011
24-11-2011
08-02-2011
01-07-2011
19-09-2011
05-12-2011
0
100
200
300
400
count


Monthly Quantity Sold
Total Quantity
250000
200000
150000
100000
50000
0
2010-12
2011-01
2011-02
2011-03
2011-04
2011-05
2011-06
2011-07
2011-08
2011-09
2011-10
2011-11
2011-12
NaT
Month


Monthly Revenue
Total Revenue
400000
300000
200000
100000
0
2010-12
2011-01
2011-02
2011-03
2011-04
2011-05
2011-06
2011-07
2011-08
2011-09
2011-10
2011-11
2011-12
NaT
Month

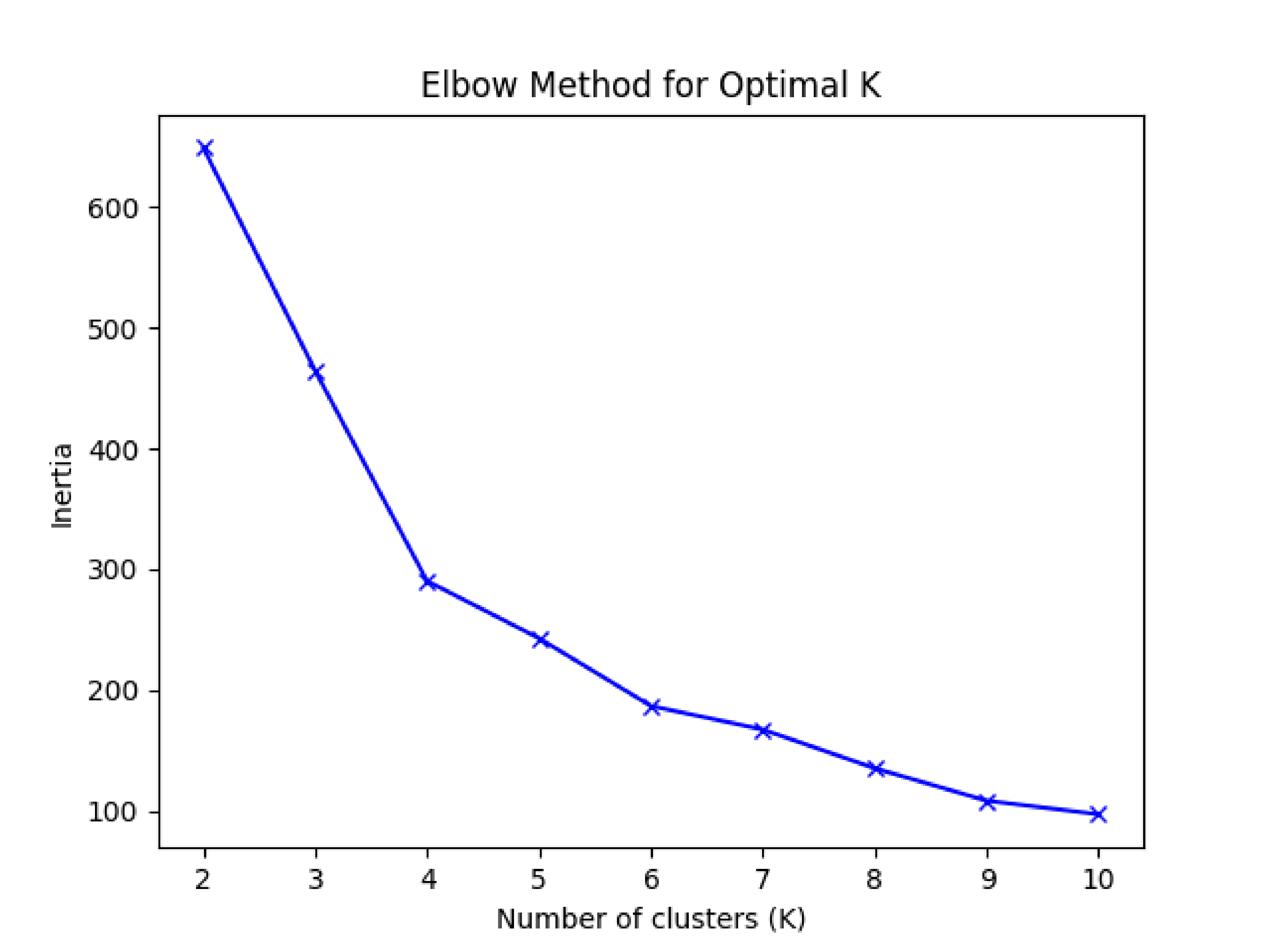

InvoiceNo
StockCode
Quantity
UnitPrice
CustomerID

05

CLUSTERS METHODOLOGY

To determine the optimal number of clusters and ensure the quality of segmentation, multiple clustering evaluation methods were applied:

1. K-Means Clustering
   - K-Means was the primary clustering algorithm used in this project.
   - It partitions the dataset into k clusters by minimizing the within-cluster sum of squares (WCSS).
   - Each data point is assigned to the nearest cluster centroid, ensuring compact and well-separated groups.
2. Elbow Method
   - This method helps determine the optimal number of clusters by plotting the WCSS against different values of k.
   - The “elbow point” on the curve indicates where adding more clusters does not significantly reduce the WCSS, suggesting the best k.
3. Silhouette Score
   - This metric evaluates how similar a data point is to its own cluster compared to other clusters.
   - The score ranges from -1 to 1: values closer to 1 indicate well-defined clusters, while values near 0 suggest overlapping clusters.
4. Calinski–Harabasz Index
   - Also known as the Variance Ratio Criterion.
   - It measures the ratio of between-cluster dispersion to within-cluster dispersion.
   - Higher values indicate better clustering performance, with compact and well-separated clusters.
5. Davies–Bouldin Index
   - This index evaluates the average similarity between each cluster and its most similar cluster.
   - Lower values indicate better clustering results, as it suggests more distinct and well-separated clusters.


Elbow Method for Optimal K
Inertia
600
500
400
300
200
100
2
3
4
5
6
7
8
9
10
Number of clusters (K)


Customer Segments (PCA Visualization)
Cluster
0
1
2
3
6
4
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0

06

RESULT AND VISUALIZATIONS

The customer segmentation project produced meaningful insights by applying clustering techniques on the cleaned dataset. The following summarizes the outcomes:

1. Clustering Results
    - Using K-Means clustering along with evaluation metrics (Elbow Method, Silhouette Score, Calinski-Harabasz, and Davies-Bouldin), the optimal number of clusters was identified.
    - The segmentation revealed distinct customer groups that differed in their purchasing behavior, order frequency, and spending capacity.
    - These clusters form the foundation for targeted marketing strategies.
2. Visualizations
3. Several plots and charts were generated to support data understanding and interpretation:
    - Exploratory Data Analysis (EDA):
        - Boxplots were created for CustomerID, InvoiceNo, Quantity, StockCode, and UnitPrice to detect outliers and unusual patterns.
        - Histograms were plotted for numerical features to understand data distribution.
        - Correlation heatmaps provided insights into relationships between variables.
        - Countplots were used for categorical attributes such as Country, Description, and InvoiceDate.
        - Monthly trends were visualized for quantity and revenue, highlighting seasonality and purchasing peaks.
        - Missing value heatmaps helped ensure the completeness of the data.
    - Clustering Visualization:
        - Scatter plots (with PCA for dimensionality reduction) were used to visualize the customer clusters in two dimensions.
        - The separation between clusters clearly illustrated the grouping effectiveness.
        - Each cluster displayed unique purchasing and spending behaviors, reinforcing the segmentation process.
4. Key Findings
    - Certain clusters represented high-value customers with large quantities and higher spending.
    - Some clusters highlighted low-frequency buyers with limited transactions.
    - Geographic segmentation (via country-based plots) revealed differences in purchasing behavior across regions.
    - Monthly analysis indicated periods of high sales activity, which can be leveraged for promotions.

Overall, the results confirmed that clustering successfully divided the customer base into meaningful groups, and the visualizations provided clarity and strong evidence for these insights.

07

CLUSTER PROFILING

Cluster profiling is the process of analyzing and interpreting the characteristics of each customer group formed after clustering. Once the dataset was segmented using K-Means, we examined the common traits, spending patterns, and demographic details within each cluster to understand their behavior.

For example, some clusters represented high-value customers who purchased frequently with higher average spending, while others included price-sensitive or low-engagement customers with smaller purchases. Certain clusters showed strong preferences for specific product categories or seasonal buying behavior, while others highlighted occasional buyers.

By profiling each cluster, we translated the numerical clustering results into meaningful customer personas, which served as the foundation for designing targeted marketing strategies. This helped in differentiating between loyal customers, new buyers, and potential churners, thereby allowing businesses to allocate resources effectively and maximize return on investment.

08 MARKETING STRATEGIES

The customer segmentation project produced meaningful insights by applying clustering techniques on the cleaned dataset. The following summarizes the outcomes:

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09 FUTURE SCOPE

- Incorporating More Features – Adding demographic attributes like age, gender, or income can provide deeper insights into customer behavior.
- Advanced Clustering Techniques – Beyond K-Means, methods like DBSCAN, Hierarchical Clustering, or Gaussian Mixture Models can be applied for more accurate segmentation.
- Real-Time Segmentation – Integrating the model into live systems would allow businesses to update customer clusters dynamically as new data arrives.
- Predictive Analytics – Future models can not only segment but also predict customer churn, lifetime value, or purchase probability.
- Integration with Marketing Systems – Linking segmentation results with CRM or marketing automation tools can directly drive targeted campaigns.
- Visualization Dashboards – Building interactive dashboards (using Power BI, Tableau, or Python libraries) can make the results more actionable for business teams.

10 RECOMMENDATIONS

- High-Value Customers (Frequent & High Spend)
- Provide loyalty programs, premium offers, and exclusive deals.
- Personalized communication to increase brand engagement.
- Price-Sensitive Customers
- Offer discounts, bundle deals, and seasonal sales.
- Highlight value-for-money promotions.
- Occasional/Seasonal Buyers
- Send reminder emails before festivals or peak seasons.
- Use targeted ads during specific time periods.
- Low-Engagement Customers
- Run re-engagement campaigns with incentives.
- Offer first-purchase discounts or free delivery to encourage repeat buying.
- Potential Growth Customers (Moderate Spend, Growing Interest)
- Upsell and cross-sell related products.
- Encourage subscriptions or memberships to build loyalty.

11 CONCLUSION

In this project, we successfully implemented customer segmentation using K-Means clustering on transactional and behavioral data. The dataset was thoroughly cleaned, pre-processed, and scaled to ensure quality inputs for clustering. Through exploratory data analysis (EDA), we identified key trends, patterns, and anomalies within customer behavior. Multiple clustering evaluation methods, including the Elbow Method, Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index, were applied to determine the optimal number of clusters.

The resulting clusters provided meaningful insights into distinct customer groups, enabling the development of targeted marketing strategies tailored to each segment. Visualizations further enhanced the interpretability of clusters, supporting decision-making for customer engagement, retention, and revenue growth.

Overall, this project demonstrates the value of data-driven segmentation in understanding customer diversity and optimizing marketing efforts. With future enhancements such as advanced clustering techniques, demographic integration, and real-time analysis, the model can evolve into a powerful tool for personalized marketing and sustainable business growth.

THANK YOU

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